Proposal	Contact PI's Surname	Initials	Application Number	Panel
Fast-Start	Ardid	AA	26-UOC-086	ESA

MARSDEN FUND PRELIMINARY RESEARCH PROPOSAL

Fast-Start Application Form

1A. RESEARCH TITLE

Understanding the Balance Between Universal and Volcano-Specific Controls on Pre-Eruptive Seismic Precursors at Data-Scarce Volcanoes

1B. IDENTIFICATION

Name (with title)	Institution	Country

Principal Investigator(s)

Dr Alberto Ardid	University of Canterbury	

Degree: ()

Thesis:

Supervisor:

Associate Investigator(s)

Dr Corentin Caudron	Université libre de Bruxelles	BELGIUM
Dr Emmanuel Caballero Leyva	Earth Sciences New Zealand	NEW ZEALAND

Mentor (Optional)

Dr Geoff Kilgour	Earth Sciences New Zealand	

1C. RESEARCH CLASSIFICATION

Fields of Research

370512 - Volcanology	40%
370601 - Applied geophysics	30%
461199 - Machine learning not elsewhere classified	30%

Keywords

Volcanic unrest; eruption precursors; volcano seismology; generalized models; fine tunning; ergodicity; ensemble learning; transferability; volcano-specific dynamics; Ruapehu; Tongariro

Socio-Economic Objectives

190403 - Geological hazards (e.g. earthquakes, landslides and volcanic activity)	60%
280107 - Expanding knowledge in the earth sciences	40%

Keywords

Volcanic unrest; geological hazards; eruption precursors; volcano seismology; earth sciences; expanding knowledge; natural hazards; seismic processes; New Zealand volcanoes

Benefit Categories

- ✓ Economic
- ✓ Environmental
- Health

New Technologies

The proposed research has the potential to support the development of new technologies.

1D. RESEARCH SUMMARY

Volcanic eruptions are rare, high-impact events, and most active volcanoes have only a small number of well-recorded eruptions. This scarcity has long limited the identification of statistically robust eruption precursors and has reinforced the view that precursory signals must be volcano-specific. Recent studies, however, show that volcanoes exhibit common pre-eruptive seismic patterns, and that models trained across many systems can successfully forecast eruptions even at data-scarce volcanoes. This project addresses a fundamental question in volcanology: how do universal, ensemble-level eruption precursors coexist with volcano-specific dynamics, and what governs departures from generalised behaviour? Building on recent advances in ensemble and transfer-learning approaches, the research will identify ergodic seismic precursors across a global multi-volcano dataset and develop principled fine-tuning strategies to isolate volcano-specific behaviour. Ruapehu and Tongariro will serve as primary case studies, allowing comparison between hydrothermally dominated unrest and predominantly magmatic systems within the broader ensemble. The scientific objective is not operational forecasting, but to use generalised models as reference frameworks to interrogate the physical meaning, limits, and interpretability of eruption precursors. The outcomes will advance fundamental understanding of volcanic unrest and the balance between shared and local processes in complex natural systems.

2. VISION MĀTAURANGA

Vision Mātauranga themes and those that apply to the proposed research.

✓ Indigenous Innovation
(Contributing to economic growth through distinctive research and development)

✓ Taiao
(Achieving environmental sustainability through iwi and hapū relationships with land and sea)

Hauora/Oranga
(Improving health and social wellbeing)

✓ Mātauranga
(Exploring indigenous knowledge and science and innovation)

Not Applicable

A brief rationale for your choice(s):

This research enhances environmental sustainability (Taiao) by improving scientific understanding of volcanic unrest in Aotearoa. Volcanoes such as Ruapehu, Tongariro and Taranaki are not only geological systems but also culturally significant landscapes. Determining whether generalized models can be refined to isolate volcano-specific signals aligns with long-term interests in understanding and living sustainably with volcanic hazards.

The project is computational and data-driven, using existing seismic datasets and machine-learning methods. No field-based research at maunga is planned. As an early-stage, discovery-focused study examining the limits of generalized forecasting models, formal community engagement is not yet practical during the initial modelling phase.

Tipene Merritt (Ngāti Kauwhata, Rangitāne, Ngā Puhi, Ngāi Te Rangi) will collaborate with the team to integrate Māori environmental perspectives and advise on appropriate framing of volcanic hazard research. He will contribute to a review of Māori perspectives and narratives associated with these volcanic landscapes, strengthening the project’s conceptual grounding and informing future engagement pathways. As research outputs emerge, he will support dissemination to Māori stakeholders in alignment with the project’s broader communication strategy.

The Mātauranga Māori component is therefore embedded in conceptual framing, environmental context, and future-oriented dissemination.

3A. ABSTRACT

Introduction. Volcanic eruptions are rare, high-impact events, yet many active volcanoes, including those in Aotearoa New Zealand, have limited (typically only one or two) well-recorded eruptions in the instrumental era[1,2,3,4,5]. This scarcity has limited our ability to identify statistically robust eruption precursors, and our ability to develop reliable forecasting models. These factors have led to a prevailing assumption that statistically robust precursors must be volcano-specific[6,7,8,9,10], and that generalisation across systems is either physically unjustified or prone to spurious correlations. However, recent studies have shown that volcanoes *do* exhibit common pre-eruptive seismic patterns[11,12,2,13,14,15,16], and data-driven approaches can outperform traditional amplitude-based metrics such as real-time seismic amplitude measurement [17,18,19,20]. Going further, recent work has shown that eruption precursors exhibit ergodicity: patterns learned from seismic data across many volcanoes can approximate the temporal evolution of an individual system [4], demonstrating that forecasting models trained on multiple volcanoes provide comparable accuracy to volcano-specific forecasting, despite having no local eruptive history in model training/calibration[4]. **Challenges**. While promising, these gains come with unresolved scientific challenges. Ensemble-trained models can obscure system-specific dynamics, and their predictive success does not necessarily imply insight into the physical processes governing unrest[3,21,5,22]. At the same time, the ability of generalized models to identify shared, transferable precursors establishes a new scientific foundation: volcanic unrest can be described in terms of ensemble-level structure rather than solely volcano-specific behaviour. A key remaining scientific gap is determining whether generalized models can be systematically refined to isolate volcano-specific signals, and whether deviations from ensemble-derived precursors reflect distinct physical processes or fundamental limits to generalisation[23]. **We ask**: How does inter-volcano heterogeneity govern the balance between ensemble-level eruptive precursors and volcano-specific dynamics, and what are the consequences for predictive skill at individual, data-scarce volcanoes? **Significance and Innovation**. We will address a foundational question in volcanology: can the preparatory processes leading to eruption can be decomposed into (i) universal components shared across systems and (ii) local components governed by individual architectures (e.g., magmatic or hydrothermal). Our innovation lies in developing the first principled framework for volcano-specific fine-tuning of generalized models, enabling discovery-level insights into system individuality even at volcanoes with extremely sparse eruptive histories, using Ruapehu as the primary case study (due to its active hydrothermal system and limited number of well-instrumented eruptions), with Tongariro and Taranaki as secondary testbeds to assess transferability across the Taupō Volcanic Zone and in contrasting tectonic settings. These volcanoes are also culturally significant landscapes in Aotearoa, reinforcing the importance of advancing fundamental understanding of their unrest dynamics. Importantly, the scientific objective is not operational forecasting; rather, it is to understand the structure of building processes and the balance between shared and local precursors. **Research Approach**. Building on previous frameworks[20,21,4,13], we will identify ergodic precursors at the ensemble, multi-volcano scale by training models on a global dataset of more than 20 volcanoes[4], spanning both phreatic and magmatic systems, using time-series feature extraction[24] and random-forest classifiers[25]. We will then develop and compare four fine-tuning strategies to isolate volcano-specific behaviour: (i) sample-weighted loss functions that emphasise local misfit during retraining; (ii) post-hoc reweighting of decision-tree outputs without retraining; (iii) stacking models that learn non-linear interactions among tree-level predictions; and (iv) a residual-learning framework in which the generalised model is frozen and a small set of new trees learns only the local correction. We will evaluate the conceptual limits of ergodicity by quantifying when and why fine-tuned models deviate from generalised ones, focusing on Ruapehu and Tongariro, which typically exhibit hydrothermally driven unrest and have limited yet well-instrumented eruptive histories. Finally, we will assess whether volcano-specific precursors revealed through fine-tuning correspond to distinct physical processes or instead reflect statistical departures from ensemble behaviour, clarifying the interpretability and physical meaning of generalised forecasting models.

3B. BENEFITS

This project will deliver fundamental advances in understanding the processes governing volcanic unrest and eruption precursors, addressing long-standing challenges in volcanology arising from the rarity of well-recorded eruptions at individual volcanoes[1,7,8]. By rigorously examining how generalised, ensemble-trained models capture both shared and volcano-specific behaviour, the research will clarify the physical meaning and limits of transferable eruption precursors[4,13,16].

Environmental and Hazard-Relevant Benefits

New Zealand hosts multiple active volcanoes characterised by hydrothermal and magmatic unrest, including Ruapehu and Tongariro, which are environmentally and culturally significant landscapes with exposure of population, infrastructure, and critical transport corridors[26-31]. Improving understanding of pre-eruptive seismic behaviour at such systems provides long-term environmental benefit by strengthening the scientific foundations underpinning volcanic hazard assessment. While not designed as an operational forecasting project, insights into the structure and interpretability of eruption precursors may inform future monitoring strategies and hazard frameworks used in Aotearoa and internationally[2,3,5,20,21].

Scientific and Knowledge Benefits

The project addresses a foundational scientific question: whether eruptive behaviour can be decomposed into universal components shared across volcanoes and local components governed by system-specific processes[4,9,23]. By evaluating when and why fine-tuned models depart from ensemble behaviour, the research will advance understanding of hydrothermal versus magmatic controls on unrest[11,12,14,30,31]. These outcomes will contribute to the broader volcanology literature on precursor interpretation, ergodicity, and the limits of generalisation in complex natural systems[4,6,15].

Capability and Strategic Benefits for New Zealand

The research will strengthen New Zealand's capability in advanced volcanic data analysis and modelling, building expertise in ensemble methods, time-series analysis, and physically informed interpretation of seismic signals[19,22,24,25]. Conducting this research in New Zealand is uniquely appropriate due to the availability of high-quality volcanic monitoring data, long-term observational records, and well-studied active systems that are globally representative yet locally distinctive[2,26–29]. The outcomes will be internationally relevant while remaining grounded in the scientific and environmental context of Aotearoa.

3B. REFERENCES

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13 Ardid, A., Dempsey, D., Caudron, C., & Cronin, S. (2022). Seismic precursors to the Whakaari 2019 phreatic eruption are transferable to other eruptions and volcanoes. *Nature Communications*, **13**, 2002.

14 Ardid, A., Dempsey, D., Caudron, C., Cronin, S. J., Miller, C. A., Melchor, I., et al. (2024). Using template matching to detect hidden fluid release episodes beneath crater lakes in Ruapehu, Copahue, and Kawah Ijen volcanoes. *Journal of Geophysical Research: Solid Earth*, **128**, e2023JB026729.

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3C. ROLES AND RESOURCES

This research will be led by a multidisciplinary team with complementary expertise in volcano seismology, data-driven modelling, and physical interpretation of volcanic unrest. The combination of ensemble modelling expertise and system-specific volcanic knowledge ensures both methodological innovation and robust geological interpretation.

Dr Alberto Ardid (Principal Investigator) will lead the project intellectually and operationally. He will be responsible for the overall research design, development and implementation of ensemble and fine-tuning modelling frameworks, and synthesis and interpretation of results. The PI will carry out the core modelling and analysis, integrate physical interpretation across case-study volcanoes, and lead all publications and dissemination arising from the project. He will work closely with a PhD student funded by this project, providing day-to-day supervision, training in data-driven volcanology and time-series analysis, and guidance in scientific writing and career development.

Dr Emmanuel Caballero (ESNZ) will contribute expertise in seismic signal analysis and data-driven methodologies applied to continuous volcanic monitoring data. His role will support methodological robustness and interpretation of seismic features in the context of real-world volcanic systems.

Dr Corentin Caudron (Université libre de Bruxelles) will provide expertise in volcano seismology, seismic attenuation, and transferable eruption precursors. He will contribute to the conceptual development of ensemble approaches and assist in interpreting whether volcano-specific deviations identified through fine-tuning reflect physical processes or statistical structure.

Dr Geoff Kilgour (ESNZ) will act as a mentor to the PI, providing guidance grounded in his extensive expertise on the volcanic processes and unrest behaviour of Ruapehu and Tongariro. His role will ensure that interpretations remain anchored in system-specific geological and hydrothermal understanding and that the research remains physically meaningful.

The collaborators will contribute through targeted scientific input, regular discussion, and co-authorship where appropriate, while responsibility for project delivery and direction rests with the PI.

Tipene Merritt (Collaborator) will provide advisory input on the integration of Māori environmental perspectives (Taiao) within the project. He will assist in developing a review component on Māori conceptualisations of volcanic landscapes and advise on culturally appropriate dissemination of research outputs as they become available. His role is advisory and dissemination-focused; no FTE is allocated.

Resources. All required data and computational resources are available. Seismic datasets for Ruapehu, Tongariro, and other New Zealand volcanoes will be accessed through established monitoring archives, alongside a global multi-volcano dataset previously compiled for ensemble modelling. The project will use established open-source tools for time-series feature extraction and machine-learning analysis, run on institutional high-performance computing facilities available to the PI. No laboratory experiments or field campaigns are required. The computational nature of the research ensures resilience to external disruptions and enables steady progress throughout the project. The host institution provides the necessary research environment, computing infrastructure, and administrative support to ensure successful project completion.

4. PERSONNEL

List the time involvement of all personnel in terms of a Full Time Equivalent (FTE). Indicate for each person whether Marsden funding will be sought for them. Give names for all personnel (except when they are as yet unknown for such people as postdoctoral fellows and postgraduate students). Please refer to the Preliminary Research Proposal Guidelines for Applicants for recommended minimum time for Principal Investigators.

Name	FTE Year 1	FTE Year 2	FTE Year 3	Marsden Funded
Principal Investigator (Contact)(s)				
Dr Alberto Ardid	0.30	0.30	0.30	✓
Associate Investigator(s)				
Dr Corentin Caudron	0.05	0.05	0.05	
Dr Emmanuel Caballero Leyva	0.05	0.05	0.05	
TOTAL	**0.40**	**0.40**	**0.40**	

New Zealand Standard Curriculum Vitae Template

Rows and columns may be expanded or reduced, but a CV must be no more than five pages in total for Part 1 and Part 2 combined. Part 2 should include information pertinent to your research proposal. Use Calibri 11 point font. Do not alter page margins. Version: December 2024

PART 1

1a. Personal details	
Title (optional)	Dr
First Name(s)	Alberto
Family Name	Ardid Segura
Iwi Affiliation, Pacific identity and/or any other as applicable	N/A
Present position	Lecturer/Assistant Professor
Organisation/Employer	University of Canterbury
Contact Address	Department of Civil and Environmental Engineering University of Canterbury Christchurch, New Zealand
Mobile	+64 27 595 4776
Email	alberto.ardid@canterbury.ac.nz
Personal website (if applicable)	LinkedIn: https://www.linkedin.com/in/alberto-ardid/ UC Profile: https://www.canterbury.ac.nz/about-uc/contact-us/staff-directory/alberto-ardid
Research identifier (if applicable)	Google Scholar: https://scholar.google.com/citations?user=Cnxcz8AAAAAJ ResearchGate: https://www.researchgate.net/profile/Alberto-Ardid-Segura-2

1b. Academic qualifications
2022, PhD in Engineering Science, Applied Geophysics, University of Auckland, New Zealand 2016, MSc in Geophysics, University of Chile, Santiago, Chile 2014, BSc in Geophysics, University of Chile, Santiago, Chile 2014, Internship, Stanford University, California, USA

1c. Professional positions held

2026-Present, Lecturer/Assistant Professor, University of Canterbury, New Zealand
2024-2026, Research Engineer, University of Canterbury, New Zealand
2022-2024, Postdoctoral Research Fellow, University of Canterbury, New Zealand

1d. Present research/professional speciality

Geophysicist and data scientist specializing in machine learning applications for natural hazard forecasting and environmental informatics. Primary research areas include:

- Volcano eruption precursors and forecasting using seismic data, magnetotelluric data, template matching, machine learning, and physics-based modeling
- Multi-hazard forecasting (wildfires, floods, avalanches) using machine learning
- Geothermal energy exploration and reservoir modeling
- Time series analysis and feature engineering for geophysical data

Expertise in developing and deploying real-time forecasting models for complex geological phenomena with applications in disaster risk reduction and early warning systems.

1e. Career break events (if applicable)

None

1f. Professional distinctions and memberships (including honours, prizes, grants, scholarships, boards, editorial and/or governance roles, etc.)

2025, UN Global Initiative AI Specialist & Chair, AI for Volcanic Hazards
2024, New Zealand Geophysics Prize, Geoscience Society of New Zealand
2024, Allianz Climate Risk Award, Selected for Compendium of Essays
2022, Dean List for Best Doctoral Theses, University of Auckland
2016, Best Master Thesis, Geophysics Department, University of Chile
2016, Best Teacher Assistant, Geophysics Department, University of Chile

1g. Total number of peer reviewed publications	Journal articles	Books	Book chapters, books edited
	10+	0	0

PART 2

2a. Research publications and dissemination

Peer-reviewed journal articles
Ardid et al., (2025) Ergodic seismic precursors and transfer learning for short term eruption forecasting at data scarce volcanoes. Nature Communications. DOI 10.1038/s41467-025-56689-x Ardid et al., (2025) Sub-hourly forecasting of fire potential using machine learning on time series of surface weather variables. International Journal of Wildland Fire 34, WF24113. Ardid et al., (2025) Teaching Flood Forecasting: Modeling River Flow Data with Machine Learning. Journal of Hydrology New Zealand. Ardid et al., (2024) Multi-Time-Scale Template Matching: Discovering Eruption Precursors Across Diverse Volcanic Settings. Seismological Research Letters, 128, e2023JB026729. Cabrera & Ardid et al., (2024) Eruption Forecasting Model for Copahue Volcano (Southern Andes) Using Seismic Data and Machine Learning. Seismological Research Letters. Ardid et al., (2023) Using template matching to detect hidden fluid release episodes beneath crater lakes in Ruapehu, Copahue and Kawah Ijen volcanoes. Journal of Geophysical Research: Solid Earth, 128, e2023JB026729. Ardid et al., (2022) Seismic precursors to the Whakaari 2019 phreatic eruption are transferable to other eruptions and volcanoes. Nature Communications, 13(1), 2002. Dempsey, Kempa-Liehr, Ardid, et al., (2022) Evaluation of short-term probabilistic eruption forecasting at Whakaari, New Zealand. Bulletin of Volcanology, 84, 91. Ardid et al., (2021) Heat Transfer through the Wairakei-Tauhara Geothermal System quantified by multi-channel data modelling. Geophysical Research Letters, 48. Ardid et al., (2021) Bayesian magnetotelluric inversion using methylene blue structural priors for imaging shallow conductors in geothermal fields. Geophysics, 86(3), 1-66. Ardid et al., (2019) Geometry of geyser plumbing inferred from ground deformation. Journal of Geophysical Research: Solid Earth, 124, 1072-1083.
Peer reviewed conference proceedings
Multiple presentations and proceedings at NZGW, NZGS, IAVCEI, JpGU, SSA, AGU, EGU, and other international conferences
Other forms of dissemination (reports for clients, technical reports, patents, popular press, public outreach, community engagement etc.)
International Media Engagement: • The Guardian (UK) (e.g., https://www.theguardian.com/world/2022/apr/20/new-zealand-scientists-find-tremor-link-that-could-predict-volcanic-eruptions) • NZ Herald, RNZ (New Zealand)

• La Tercera, Cooperativa (Chile)
• Cosmos Magazine (Australia)

Conference Keynote Talks:
• FireNZ 2025, UFBA 2025, IDI-UNICyT 2025

Invited Speaker:
• IAVCEI 2025, 2023; JpGU 2022; SSA 2024

Community Engagement:
• UN Global Initiative AI Specialist & Chair for AI for Volcanic Hazards (2025-Present)
• Free Workshop lecturer: Machine Learning and Neural Networks, Python for Geoscientists, LaTeX for Beginners

Proposal	Contact PI's Surname	Initials	Application Number	Panel
Fast-Start	Ardid	AA	26-UOC-086	ESA

New Zealand Standard Curriculum Vitae Template

Rows and columns may be expanded or reduced, but a CV must be no more than five pages in total for Part 1 and Part 2 combined. Part 2 should include information pertinent to your research proposal. Use Calibri 11 point font. Do not alter page margins. Instructions in italics should be deleted before you submit your CV. Version: December 2024

PART 1

1a. Personal details	
Title (optional)	Dr
First Name(s)	Corentin Jules
Family Name	Caudron
Iwi Affiliation, Pacific identity and/or any other as applicable	
Present position	Associate Professor
Organisation/Employer	Université libre de Bruxelles, Belgium
Contact Address	Département des Géosciences Environnement et Société
	Av. Franklin Roosevelt 50, Bruxelles, 1050
Mobile	+32485341293
Email	corentin.caudron@ulb.be
Personal website (if applicable)	https://gtime.ulb.be/
Research identifier (if applicable)	https://orcid.org/0000-0002-3748-0007

1b. Academic qualifications

2013, Ph.D. in Science, Université Libre de Bruxelles (Belgium)
2009, M.Sc., Earth sciences, Université Libre de Bruxelles (Belgium)
2007, B.Sc., Earth Sciences, Université Libre de Bruxelles (Belgium)

1c. Professional positions held

2021, Associate Professor, ULB, Brussels, Belgium
2018-2021, Chargé de Recherche IRD, ISTerre, Grenoble, France
2017-2018, Doctor-Assistant, University of Ghent, Belgium
2016-2017 FNRS postdoctoral Fellow, Université Libre de Bruxelles, Belgium
2015-2016, Postdoctoral Fellow, University of Cambridge, UK
2013-2015, Postdoctoral Fellow, Earth Observatory of Singapore, Magma Transport Dynamics - Nanyang Technological University, Singapore.

1d. Present research/professional speciality

My research focusses on volcanoes and more specifically seeks to understand magmatic hydrothermal systems (so-called 'wet volcanoes') using seismo-acoustic methodologies focussed on continuous seismicity (ambient seismic noise).

Keywords: Hydrothermal systems and gas-driven eruptions – Magma dynamics – Seismic noise and tremor - Volcanic Lakes – Infrasound

1e. Career break events (if applicable)

List any significant interruptions to your career (e.g. sickness, parental leave). Delete and start typing here.

1f. Professional distinctions and memberships (including honours, prizes, grants, scholarships, boards, editorial and/or governance roles, etc.)

Delete and start typing here. List in reverse date order. Start each professional distinction on a new line as per the example:
e.g. Year / year-year, distinction.

2024-now; SAC (Scientific Advisory Committee, funded by the British government) for Montserrat volcano
2026-now; Appointed member of the Collegium of the Belgian Academy (Classe Sciences, since 2026)
2025-now; Appointed effective member (IAVCEI) of the Belgian National
Committee for Geodesy and Geophysics (BNCGG)
2026-now; Co-chair of the IAVCEI/IASPEI Commission of Volcano Seismology and Acoustics
2024-now; I have also been selected to serve for the EPOS-France Scientific Committee for the Volcanology

2025; Editor IAVCEI Book series Advances in Volcanology and Active Volcanoes of the World: "Modern Volcano Monitoring"
2024, Guest Invited Editor for a Special Volume 'Multiscale Approach and Integrated Multiplatform Techniques to Study the Volcanic and Geothermal Environments' (Frontiers in Earth Sciences)
2019 – 2023, Co-Leader of the IAVCEI Commission on Volcanic Lakes (https://iavcei-cvl.org/)
2021 – 2023, Chair of Focus Group (Volcanic Eruption Forecasting) on Artificial Intelligence for Natural Disaster Management (United Nations)
2022, Guest Invited Editor for a Special Volume 'Volcanic Lake Dynamics and Related Hazards" (Frontiers in Earth Sciences)
2019-2021, Associate Editor for Journal of Geophysical Research-Solid Earth
2018, Prize Baron Van Ertborn (best publication in Geosciences) – Académie Royale des Sciences (Belgium)
2017, Guest Invited editor for a Special Volume "Geochemistry and Geophysics of Active Volcanic Lakes" (Geological Society of London)
2010, Best presentation award: PhD Student day of ENVITAM school (Namur, Belgium)

1g. Total number of *peer reviewed* publications	Journal articles	Books	Book chapters, books edited	Conference proceedings
	73		4,2	>150

Proposal	Contact PI's Surname	Initials	Application Number	Panel
Fast-Start	Ardid	AA	26-UOC-086	ESA

PART 2

2a. Research publications and dissemination

Expand/reduce the following table as needed, listing publications relevant to your proposal. List in reverse date order. Bold your name in lists of authors.

Peer-reviewed journal articles

Lesage, P., Caudron, C. and Yates, A., 2025. Coda Wave Interferometry for Volcano Monitoring. In Modern Volcano Monitoring (pp. 147-188). Cham: Springer Nature Switzerland.

Jousset, P., Currenti, G., Murphy, S., Eibl, E.P., Caudron, C., Retailleau, L., Wollin, C., Klaasen, S., Nishimura, T., Fichtner, A. and Spica, Z., 2025. Fiber optic sensing for volcano monitoring and imaging volcanic processes. In *Modern volcano monitoring* (pp. 509-567). Cham: Springer Nature Switzerland.

Reiss, M.C., Caudron, C., Hering, P. and Roman, D., 2025. Tremor signals reveal the structure and dynamics of the Oldoinyo Lengai magmatic plumbing system. Communications Earth & Environment, 6(1), p.781.

Roche, B., Caudron, C., Van der Laat, L., De Moor, J.M., Avard, G., Pacheco, J., Bakkar, H., Govoorts, J. and Rodriguez, A., 2025. First hydroacoustic recording of ebullition events in an active volcanic lake–Poás, Costa Rica. Volcanica, 8(1), pp.225-239.

Yates, A., Caudron, C., Mordret, A., Lesage, P., Cannata, A., Cannavo, F., Lecocq, T., Pinel, V. and Zaccarelli, L., 2025. Precursory velocity changes prior to the 2019 paroxysms at Stromboli volcano, Italy, from coda wave interferometry. *Volcanica*, *8*(1), pp.203-223.

Ardid, A., Dempsey, D., Caudron, C., Cronin, S., Kennedy, B., Girona, T., Roman, D., Miller, C., Potter, S., Lamb, O.D. and Martanto, A., 2025. Ergodic seismic precursors and transfer learning for short term eruption forecasting at data scarce volcanoes. *Nature communications*, *16*(1), p.1758.

Soubestre, J., Caudron, C., Melnik, O., Lecocq, T., Jaupart, C., Shapiro, N.M., Journeau, C., Çubuk-Sabuncu, Y. and Jónsdóttir, K., 2025. Dynamics of the 2021 Fagradalsfjall eruption (Iceland) revealed by volcanic tremor patterns. *Journal of Geophysical Research: Solid Earth*, *130*(2), p.e2024JB029380.

Steinke, B., Jolly, A.D., Girona, T., Caudron, C., Bramwell, L.A., Cronin, S.J., Illsley-Kemp, F. and Hughes, E.C., 2024. Dynamics and Detection of Pulsed Tremor at Whakaari (White Island), Aotearoa New Zealand. *Geophysical Research Letters*, *51*(20), p.e2024GL110447.

Svennevig, K., Hicks, S.P., Forbriger, T., Lecocq, T., Widmer-Schnidrig, R., Mangeney, A., Hibert, C., Korsgaard, N.J., Lucas, A., Satriano, C. and Anthony, R.E.,..., C. Caudron,... 2024. A rockslide-generated tsunami in a Greenland fjord rang Earth for 9 days. Science, 385(6714), pp.1196-1205.

Yates, A.S., Caudron, C., Mordret, A., Lesage, P., Pinel, V., Lecocq, T., Miller, C.A., Lamb, O.D. and Fournier, N., 2024. Seasonal snow cycles and their possible influence on seismic velocity changes and eruptive activity at Ruapehu volcano, New Zealand. *Journal of Geophysical Research: Solid Earth*, *129*(12), p.e2024JB029568.

Caudron, C., Miao, Y., Spica, Z.J., Wollin, C., Haberland, C., Jousset, P., Yates, A., Vandemeulebrouck, J., Schmidt, B., Krawczyk, C. and Dahm, T., 2024. Monitoring underwater volcano degassing using fiber-optic sensing. Scientific Reports, 14(1), p.3128.

Contreras-Arratia, R., Chouet, B., Caudron, C., Nath, N., Balchan, A., Madoo, F. and Ramsingh, H., 2024. Tracking seismicity in an underfunded institution: The case of La Soufrière St Vincent volcanic eruption 2020–2021. Journal of Volcanology and Geothermal Research, p.107990.

Muzellec*, T., Lesage, P., Caudron, C. and Got, J.L., 2023. Migration of Mechanical Perturbations Estimated by Seismic Coda Wave Interferometry During the 2018 Pre-Eruptive Period at Kīlauea Volcano, Hawaii. Journal of Geophysical Research: Solid Earth, 128(8), p.e2022JB026224.

Ardid, A., Dempsey, D., Caudron, C., Cronin, S.J., Miller, C.A., Melchor, I., Syahbana, D. and Kennedy, B., 2023. Using template matching to detect hidden fluid release episodes beneath crater lakes in Ruapehu, Copahue and Kawah Ijen volcanoes. Journal of Geophysical Research: Solid Earth, 128(10), p.21.

Machacca, R. *, Lesage, P., Tavera, H., Pesicek, J.D., Caudron, C., Torres, J.L., Puma, N., Vargas, K., Lazarte, I., Rivera, M. and Burgisser, A., 2023. The 2013–2020 seismic activity at Sabancaya Volcano (Peru): Long lasting unrest and eruption. Journal of Volcanology and Geothermal Research, 435, p.107767.

Proposal	Contact PI's Surname	Initials	Application Number	Panel
Fast-Start	Ardid	AA	26-UOC-086	ESA

Yates*, A., Caudron, C., Lesage, P., Mordret, A., Lecocq, T. and Soubestre, J., 2023. Assessing similarity in continuous seismic cross-correlation functions using hierarchical clustering: application to Ruapehu and Piton de la Fournaise volcanoes. Geophysical Journal International, 233(1), pp.472-489.

Subira, J. *, Barrière, J., Caudron, C., Hubert-Ferrari, A., Oth, A., Smets, B., d'Oreye, N. and Kervyn, F., 2023. Detecting sources of shallow tremor at neighboring volcanoes in the Virunga Volcanic Province using seismic amplitude ratio analysis (SARA). Bulletin of Volcanology, 85(5), p.27.

Caudron, C., Aoki, Y., Lecocq, T., De Plaen, R., Soubestre, J., Mordret, A., Seydoux, L. and Terakawa, T., 2022. Hidden pressurized fluids prior to the 2014 phreatic eruption at Mt Ontake. Nature Communications, 13(1), pp.1-9.

Montanaro, C., Mick, E., Salas-Navarro, J., Caudron, C., Cronin, S.J., de Moor, J.M., Scheu, B., Stix, J. and Strehlow, K., 2022. Phreatic and Hydrothermal Eruptions: From Overlooked to Looking Over. Bulletin of Volcanology, 84(6), pp.1-16.

Matoza, R. S., Fee, D., Assink, J. D., Iezzi, A. M., Green, D. N., Kim, K., C. Caudron,... & Wilson, D. C. 2022. Atmospheric waves and global seismoacoustic observations of the January 2022 Hunga eruption, Tonga. Science, eeeeeeeeeee.

Smittarello, D., Smets, B., Barrière, J., Michellier, C., Oth, A., Shreve, T., C. Caudron ... & Syavulisembo Muhindo, A. 2022. Precursor-free eruption triggered by edifice rupture at Nyiragongo volcano. Nature, 609(7925), 83-88.

Caudron, C., Vandemeulebrouck, J. and Sohn, R.A., 2022. Turbulence-induced bubble nucleation in hydrothermal fluids beneath Yellowstone Lake. *Communications Earth & Environment*, *3*(1), pp.1-6.

Caudron, C., Soubestre, J., Lecocq, T., White, R.S., Brandsdóttir, B. and Krischer, L., 2022. Insights into the dynamics of the 2010 Eyjafjallajökull eruption using seismic interferometry and network covariance matrix analyses. Earth and Planetary Science Letters, 585, p.117502.

Ardid, A., Dempsey, D., Caudron, C., Cronin, S. 2022. Seismic precursors to the Whakaari 2019 phreatic eruption are transferable to other eruptions and volcanoes. *Nat Commun* 13, 2002.

Caudron, C., Girona, T., Jolly, A., Christenson, B., Savage, M.K., Carniel, R., Lecocq, T., Kennedy, B., Lokmer, I., Yates, A. and Hamling, I., 2021. A quest for unrest in multiparameter observations at Whakaari/White Island volcano, New Zealand 2007–2018. Earth, Planets and Space, 73(1), pp.1-21. (Invited Frontiers Letter)

Nowé Sylvain, Thomas Lecocq, Corentin Caudron, Kristín Jónsdóttir and Frank Pattyn 2021. "Permanent, seasonal and episodic seismic sources around Vatnajokull, Iceland from the analysis of correlograms", Volcanica, 4(2).

Cubuk-Sabuncu, Y., Jónsdóttir, K., Caudron, C., Lecocq, T., Parks, M.M., Geirsson, H. and Mordret, A. 2021. Temporal Seismic Velocity Changes During the 2020 Rapid Inflation at Mt. Þorbjörn-Svartsengi, Iceland, Using Seismic Ambient Noise. Geophysical Research Letters, p.e2020GL092265.

Bernard, A., Villacorte, E., Maussen, K., Caudron, C., Robic, J., Maximo, R.P., Rebadulla, R., Bornas, M., Solidum Jr., R., 2021, Carbon dioxide in Taal volcanic lake: a simple gasometer for volcano monitoring, Geophysical Research Letters, 47, e2020GL090884

Lecocq, T., Hicks, S.P., Van Noten, K., van Wijk, K., Koelemeijer, P., De Plaen, R.S., Massin, F., Hillers, G.,..., Caudron C.,..., 2020. Global quieting of high-frequency seismic noise due to COVID-19 pandemic lockdown measures. Science.

Ashton F. Flinders, Caudron, C. Ingrid A. Johanson, Taka'aki Taira, Brian Shiro, Matthew Haney, 2020. Seismic velocity variations associated with the 2018 Lower East Rift Zone eruption of Kīlauea, Hawai'i, Special Volume on the 2018 Kīlauea eruption, Hawai'i, Bulletin of Volcanology, 82, p.47. IF2016: 2.58.

Jolly, A., Caudron, C., Girona, T., Christenson, B., Carniel, R. 2020. 'Silent' Dome Emplacement into a Wet Volcano: Observations from an Effusive Eruption at White Island (Whakaari), New Zealand in Late 2012. Geosciences, 10, 142, Special Issue, Exploring and Modeling the Magma-Hydrothermal Regime, Geosciences 10, no. 4 (2020): 142.

Métaxian, J. P., Santoso, A. B., Caudron, C., Cholik, N., Labonne, C., Poiata, Beauducel, F., Monteiller, V., Fahmi, A.A., Rizal, M.H., Nandaka, I.M.A, 2020. Migration of seismic activity associated with phreatic eruption at Merapi volcano, Indonesia, Journal of Volcanology and Geothermal Research, 106795.

New Zealand Standard Curriculum Vitae Template

Rows and columns may be expanded or reduced, but a CV must be no more than five pages in total for Part 1 and Part 2 combined. Part 2 should include information pertinent to your research proposal. Use Calibri 11 point font. Do not alter page margins. Instructions in italics should be deleted before you submit your CV. Version: December 2024

PART 1

1a. Personal details	
Title (optional)	Dr.
First Name(s)	Emmanuel
Family Name	Caballero Leyva
Iwi Affiliation, Pacific identity and/or any other as applicable	
Present position	Seismologist
Organisation/Employer	Earth Sciences New Zealand
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	PO Box 30-368, Lower Hutt 5040
Mobile	21 16 73 477
Email	e.caballero@gns.cri.nz
Personal website (if applicable)	*e.g., Personal blog, LinkedIn profile, employee page on your employer's website*
Research identifier (if applicable)	*https://orcid.org/0000-0002-2514-0670* *https://scholar.google.com/citations?user=SdvIXvsAAAAJ&hl=es*

1b. Academic qualifications

2022: Ph.D. Earth Sciences (University of Strasbourg, France)

2017: M.Sci. Seismology (National Autonomous University of Mexico, UNAM, Mexico)

2015: Geophysical Engineering degree (National Autonomous University of Mexico, UNAM, Mexico)

1c. Professional positions held

2024 – present: Seismologist, Earth Sciences New Zealand (ESNZ), New Zealand

2022 – 2024: Post-doctoral researcher, Institut of Sciences de la Terre (ISTerre), France

2020 – 2022: Research Engineering, EOST/CNRS, University of Strasbourg, France

2017 – 2018: Research Assistant, Institute of Engineering, National Autonomous University of Mexico, UNAM, Mexico

1d. Present research/professional speciality

Dr Emmanuel Caballero is a seismologist focused on natural hazard mitigation through the analysis of seismic wavefields and the development of innovative, data-driven methodologies. His expertise lies in integrating diverse geophysical datasets to characterise complex Earth processes and to develop tools for understanding and monitoring hazardous phenomena. He has conducted research across a wide range of tectonic settings, including Mexico, Chile, France, and New Zealand. In volcanic environments, his work at

Piton de la Fournaise has focused on the analysis of continuous seismic signals to identify and interpret source processes at depth, contributing to improved understanding of volcanic dynamics.

1e. Career break events (if applicable)

List any significant interruptions to your career (e.g. sickness, parental leave). Delete and start typing here.

1f. Professional distinctions and memberships (including honours, prizes, grants, scholarships, boards, editorial and/or governance roles, etc.)

2025–2030 | Associate Investigator | MBIE Endeavour Research Programme | Next-generation early warning: Forecasting tsunami and multi-hazard impacts as local earthquakes strike.

2025 | Primary Investigator | Strategic Development Funding (SDF), Internal ESNZ grant | Unveiling earthquake complexity using multi-data approaches.

2023 – 2024 | Participant | SEISMAZE ERC | Data-intensive analysis of seismic tremors and long period events: a new paradigm for understanding transient deformation processes in active geological systems.

2020-2022 | Participant | PRESEISMIC ERC | Exploring the nucleation of large earthquakes: cascading and unpredictable or slowly driven and forecastable.

2019 – 2021: PhD scholarship, Science and Technology National Council (CONACYT), Mexico.

2015 – 2017: Master’s scholarship, Science and Technology National Council (CONACYT).

2017: Honorific mention, master’s thesis defense, UNAM, Mexico.

2024 – present: Member of the internal Earthquake Expert Panel (EEP), ESNZ.

2024 – present: Member of the Geoscience Society of New Zealand, GSNZ.

2023 – present: Member of the European Geosciences Union, EGU.

1g. Total number of *peer reviewed* publications	Journal articles	Books	Book chapters, books edited	Conference proceedings
	5			10

Proposal	Contact PI's Surname	Initials	Application Number	Panel
Fast-Start	Ardid	AA	26-UOC-086	ESA

PART 2

2a. Research publications and dissemination

Expand/reduce the following table as needed, listing publications relevant to your proposal. List in reverse date order. Bold your name in lists of authors.

Peer-reviewed journal articles

- Caballero E., Andrews, J., D'Anastasio, E., Crowell, B., Ulberg, C., Solares, M., Melgar, D., Howell, A., Kaiser, A., and Fry, B. (2026). GFAST-NZ: Rapid geodetic inversion for finite faults in New Zealand. *Submitted to Seismological Research Letters*
- Solares Colon, M. M., Melgar, D., Howell, A., Crowell, B., D'Anastasio, E., Caballero, E., & Fry, B. (2025). Using ruptures from an earthquake cycle simulator to test geodetic early warning system performance. Seismica, 4(2). https://doi.org/10.26443/seismica.v4i2.1769
- Caballero, E., Duputel, Z., Twardzik, C., Rivera, L., Klein, E., Jiang, J., ... & Simons, M. (2023). Revisiting the 2015 M w= 8.3 Illapel earthquake: unveiling complex fault slip properties using Bayesian inversion. Geophysical Journal International, 235(3), 2828-2845.
- Caballero, E., Chounet, A., Duputel, Z., Jara, J., Twardzik, C., & Jolivet, R. (2021). Seismic and aseismic fault slip during the initiation phase of the 2017 MW= 6.9 Valparaíso earthquake. Geophysical research letters, 48(6), e2020GL091916.
- Cruz-Atienza, V. M., Husker, A., Legrand, D., Caballero, E., & Kostoglodov, V. (2015). Nonvolcanic tremor locations and mechanisms in Guerrero, Mexico, from energy-based and particle motion polarization analysis. Journal of Geophysical Research: Solid Earth, 120(1), 275-289.

Peer reviewed books

Peer reviewed book chapters, books edited

Peer reviewed conference proceedings

- Caballero, E., Andrews, J., D'Anastasio, E., Crowell, B., Ulberg, C., Melgar, D., Solares, M., Howell, A., Kaiser, A., and Fry, B. GFAST-NZ: Rapid geodetic inversion for finite faults in New Zealand, 12th ACES International Workshop 2025, Taipei, Taiwan, 3–7 Nov 2025, 2025.
- Caballero, E., Andrews, J., D'Anastasio, E., Crowell, B., Ulberg, C., Solares, M., Melgar, D., Howell, A., Kaiser, A., and Fry, B. Preparing New Zealand for natural hazards: Implementation of real-time GNSS and G-FAST for fast earthquake characterization, GSNZ Annual Conference 2024, Dunedin, New Zealand, 25–29 Nov 2024, 2024.
- Caballero, E., Shapiro, N., Journeau, C., Seydoux, L., Soubestre, J., and Barajas, A. Long-term evolution of continuous seismic signals at Piton de la Fournaise volcano inferred from the network covariance matrix, EGU General Assembly 2024, Vienna, Austria, 14–19 Apr 2023, EGU24-6379, https://doi.org/10.5194/egusphere-egu24-6379, 2024.
- Caballero, E., Shapiro, N., Journeau, C., Seydoux, L., Soubestre, J., and Barajas, A.: Long-term evolution of the spectral content of continuous seismo-volcanic signals from a network-based analysis, EGU General Assembly 2023, Vienna, Austria, 24–28 Apr 2023, EGU23-2724, https://doi.org/10.5194/egusphere-egu23-2724, 2023.
- Duputel, Z., Caballero, E., Twardzik, C., Rivera, L., Klein, E., Jiang, J., Liang, C., Zhu, L., Jolivet, R., Fielding, E., and Simons, M.: Revisiting the 2015 Mw=8.3 Illapel earthquake: Unveiling complex fault

slip properties using Bayesian inversion., EGU General Assembly 2023, Vienna, Austria, 24–28 Apr 2023, EGU23-6666, https://doi.org/10.5194/egusphere-egu23-6666, 2023.

- Caballero E, Z. Duputel, C. Twardzik, E. Klein, H. Aochi, J. Jiang, C. Liang, L. Zhu, R. Jolivet, E. Fielding, and M. Simons. Revisiting the 2015 Mw=8.3 Illapel earthquake. From kinematic rupture inversion to rupture dynamics. American Geophysical Union (AGU), New Orleans, USA, abstract S55D-0179, 2021.
- Caballero E, Z. Duputel, H. Aochi, H. Bhat, N. Brantut, R. Jolivet. Insight on rupture dynamics from probabilistic fault models of Chilean large earthquakes. American Geophysical Union (AGU), San Francisco, USA, abstract S037-0007, 2020.
- Caballero E, A. Chounet, Z. Duputel, J. Jara, R. Jolivet. The pre-seismic phase of the 2017 Valparaiso Mw 6.9 earthquake : assessment of aseismic to seismic partitioning with reliable uncertainty estimates. American Geophysical Union (AGU), San Francisco, USA, abstract S21F-0582, 2019.
- Caballero E, V. M. Cruz-Atienza. Moment tensor inversion of tectonic tremors in the Guerrero subduction zone. InternationalAssociation of Geodesy and international Association of Seismology and Physics of the Earth's Interior (IAG-IASPEI), Kobe, Japan, Abstract U000739, 2017.
- Caballero E, V. M. Cruz-Atienza. Moment tensor inversion of tectonic tremors in the Guerrero subduction zone. Mexican Geophysical Union (UGM), Puerto Vallarta, México, Geos, Vol. 36, 2016.

Other forms of dissemination (reports for clients, technical reports, patents, popular press, public outreach, community engagement etc.)

6. OTHER FUNDING

List of other funding organisations to whom you have sought or received a grant for this application.

No other funding applications listed.